

Building Animation Classes

The basic idea behind an animation class is relatively straightforward. The client will provide an ordered set of images and an amount of time. The client code will tell the animation when to start playing and will periodically tell the animation to update itself. The images in the animation will be displayed in order, each for the given amount of time.

SimpleAnimation Class

The general technique is to begin by loading the complete set of images, storing them in a list, and displaying the first image. When the client tells the animation to start, the animation begins tracking time. Each time the object is told to update itself, our code checks to see if the specified amount of time has passed and, if so, displays the next image in the sequence. When the animation is finished, we display the first image again.

Creating the Class

[Listing 14-1](#) contains the code of a `SimpleAnimation` class, which handles an animation made up of separate image files. To keep things clearly organized, I strongly recommend that you place all the image files associated with an animation in a subfolder inside an *images* folder inside your project folder. The examples given here will use this structure, and the associated art and main code are available with the rest of the book's resources.

File: SimpleAnimation/SimpleAnimation.py

```
# SimpleAnimation class

import pygame
import time

class SimpleAnimation():
    def __init__(self, window, loc, picPaths durationPerImage): ❶
        self.window = window
        self.loc = loc
        self.imagesList = []
```